



Proposed CEA Link — December 2026

The supplied 24-line design with its two gate failures repaired and the rotation re-searched — every duty, every headcount and the whole coverage curve unchanged

Prepared 18 September 2026 · re-rendered 24 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules (Marylbone Roster v24.22) · figures on this page are computed, not typed

PROPOSAL

Fifteen Turns Repaired

The supplied design, repaired and re-searched

CODE FT-24-R21

FINGERPRINT b76bf9e1

BUILT FROM duty table — · seed — · 24 lines · 4 cover weeks

LINEAGE Supplied as a Word table, then REPAIRED and re-searched here — one duty shortened by an hour to clear both gates, and the rotation searched with every duty, headcount and coverage hour held fixed. Compare with the design as supplied ("Fifteen Turns", FT-24-EXT · 9a028392), and with "Same Turns" (ST-24-B7 · d15e1b74) and "By the Book" (BB-24-D7 · 0f14abce), both built by the proposal search.

✓ **24 lines** designed

✓ **35h 00m** a week — exactly the contract

✓ **0** hard-limit breaches

✓ **0** rests under 12h

⚠ **1 fatigue factor** present (today: 5)

4 of 15

shift times are today's

11 times are new; the rest are turns people already work

6 days

longest run of worked days

today's link reaches 9; Chiltern's limit is 13

16 of 20

working weeks are one turn

same clock time all week, as 7 of today's 16 are

6 in 24

full weekends off

25% of the rotation, against 20% today (4 in 20)

4 cover weeks

lines 6, 12, 18, 24

gaps of 6, 6, 6, 6 around the wheel — a perfect spread, kept through the search

3h 27m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24 rather than 20. The brief was to keep the **feel** of the link people work today — a week is one turn, rest days come in pairs, four whole cover weeks spread round the rotation, and the same 06:20 / 08:00 / 14:00 / 15:15 clock times — while meeting the staffing shape agreed for December 2026 (four to open every day, three through to the close, four on a Saturday, five still on at 22:00, fourteen on a Saturday, ten on a Sunday) and every rule the link is assessed against: Chiltern's 13-day limit, twelve hours between duties, the exact 35-hour contracted week, and the ORR's good-practice fatigue factors.

It was not drawn by this app. It arrived as a Word table — a proposal from outside the workspace — and was read straight into the app's own shape, then judged by the same modules that judge a design the workspace generated itself: runDesignChecks, weeklyHours, assessHardLimits, assessFatigue, scoreOrder and calcHourlyCoverage. There is no search behind it and nothing to reproduce; what can be checked is every figure on these pages, each computed from the cells on page 7. Paste those cells into Links → Import and the workspace restates all of it.

What it is not. Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December 2026 headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 6 and 7 say so where it applies. It clears every rule the tool can check, which is the floor for being worth discussing.

BEFORE THE FIGURES

The decision this sheet supports

What is being decided, which criteria are hard and which are the room's, how much each figure can bear, and what no page here can tell you. Read this before the tiles.

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What this is
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A week compared with today
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The ORR fatigue factors
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The decision it supports (this page)
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People on duty, hour by hour
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Where it came from
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The 24-line grid, with the words explained
- 6

Hard limits and the December 2026 figures
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The grid, ready to paste into the app

The question

Should the link be built on the supplied fifteen-turn table, accepting 13 on a Saturday and 10 on a Sunday against the December 2026 staffing shape, for a table with the fewest distinct times here?

Fifteen Turns Repaired is read against **Same Turns (ST-24-B7)**, which keeps today's times at the same fatigue count — not against every design in the folder.

The criteria, in three tiers

HARD — A DESIGN MEETS THESE OR CANNOT BE RUN

- ✓ Twelve hours between duties — met, tightest 14h 05m
- ✓ Chiltern's 13-day limit — longest run 6 days
- ✓ The 35-hour contract, exactly — 35h 00m

SOFT — THE ROOM WEIGHS THESE AGAINST EACH OTHER

- December 2026 headcounts met: **1 of 4** (today 0 of 4)
- Fatigue factors present: **1** (today 5)
- Demand fit weekday · Sat · Sun: **30.7 · 31.6 · 43** (today 44.7 · 45.7 · 65.9; lower is closer)
- Times worked today: **4 of 15** · full weekends off: **6** in 24 (today 4 in 20)

PREFERENCE — STAFF'S TO STATE, NOT THE TOOL'S

- Early against late: earlies 7h 00m–8h 30m, lates 7h 00m–8h 30m
- 11 of 15 clock times are new — the duty table on page 4 lists them
- The longest duty, 8h 30m, and who holds it
- Cover weeks at lines 6, 12, 18, 24

How much to trust each figure

FIGURE ON THESE PAGES	WHERE IT COMES FROM	WEIGHT
0 rests under 12h · 35h 00m contract · 13/10 at the weekend	Computed from the cells; re-computed by the workspace on import	Firm
Longest run 6 days, inside 13	Computed; the 13-day limit's policy citation is still outstanding	Firm figure , unconfirmed threshold
Four at the open, three to close, 14 and 10 at the weekend	Owner-relayed December 2026 headcounts with no document behind them (class C)	Provisional
1 fatigue factor present	The ORR's 2021 list, which it states are not limits; 4 factors still "definition to confirm"	Advisory only — never pass or fail
Demand fit 30.7 · 31.6 · 43	Measured timetable, one share-based measure on one December 2026 timetable curve	Indicative
4 of 15 times are today's	Computed against the live 20-line link	Firm , but a proxy for acceptability

What no page here can tell you

- Whether the people who would work them accept a table in which 11 of 15 times are new (page 4 lists them) — familiarity is measured as "times worked today", which is a proxy for acceptability, not acceptability.
- Whether the roster clerk splits a cover week day-on-day-off. It is the one thing that moves the 48-hour-break figure (10 as a block, 11 split), and the link does not decide it.
- How leave and sickness cover land on a 24-line link with four cover weeks (lines 6, 12, 18, 24); nothing here models absence.
- Whether the December 2026 timetable demand curve holds on Wembley event days — the Saturday weighting approximates them, and the curve is a measured timetable, not a footfall count.

Before it is frozen

Lines 1, 2, 3, 4, 7 hold the times nobody works today; the people who would work those weeks read them first and say in one line each what they would change. The roster office answers the cover-week question. Someone confirms the source of the 13-day limit. Only then does the factor count on page 7 mean anything, and only then is this sheet more than an aid to a conversation.

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON-SAT hours	ALL WEEK hours	DAYS WORKED Mon-Sat	TURNS Mon-Fri
1	07:15 15:45	06:20 13:50	RD	RD	06:20 13:50	07:00 15:00	06:20 14:20	31h 00m	39h 30m	4	2
2	07:15 15:45	06:20 13:20	RD	06:20 13:20	06:20 13:20	06:20 13:20	06:20 14:50	36h 30m	45h 00m	5	1
3	RD	RD	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	15:00 23:00	38h 00m	38h 00m	5	1
4	RD	06:20 13:50	06:20 13:50	RD	RD	06:20 13:50	07:00 15:00	30h 30m	30h 30m	4	1
5	07:15 15:45	06:20 13:50	06:20 13:50	RD	RD	06:20 13:50	07:00 15:00	30h 30m	39h 00m	4	1
6	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
7	15:25 23:25	16:25 23:55	RD	16:25 23:55	16:25 23:55	16:25 23:55	RD	30h 00m	38h 00m	4	1
8	RD	14:00 22:30	14:00 22:30	14:00 22:30	14:00 22:30	15:00 23:00	RD	42h 00m	42h 00m	5	2
9	RD	13:00 21:30	13:00 21:30	13:00 21:30	13:00 21:30	RD	15:55 23:55	42h 00m	42h 00m	5	1
10	14:00 22:00	14:00 22:30	14:00 22:30	14:00 22:30	RD	RD	15:55 23:55	33h 30m	41h 30m	4	1
11	15:25 23:25	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	RD	RD	30h 00m	38h 00m	4	1
12	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
13	14:00 22:00	13:00 21:30	RD	13:00 21:30	14:00 22:30	14:00 22:30	RD	34h 00m	42h 00m	4	2
14	RD	07:00 15:00	07:00 15:00	07:00 15:00	07:00 15:00	07:00 15:00	RD	40h 00m	40h 00m	5	1
15	RD	06:20 14:20	06:20 13:50	06:20 13:50	06:20 13:50	RD	RD	30h 00m	30h 00m	4	3
16	RD	07:00 15:00	07:00 15:00	07:00 15:00	07:00 15:00	RD	06:20 14:20	40h 00m	40h 00m	5	1
17	15:25 23:25	16:25 23:55	16:25 23:55	RD	RD	16:25 23:55	15:55 23:55	30h 30m	38h 30m	4	1
18	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
19	10:00 18:30	RD	RD	13:00 21:30	13:00 21:30	13:00 21:30	13:00 21:30	34h 00m	42h 30m	4	1
20	RD	RD	13:00 21:30	13:00 21:30	13:00 21:30	13:00 21:30	13:00 21:30	42h 30m	42h 30m	5	1
21	RD	RD	07:00 15:00	07:00 15:00	07:00 15:00	07:00 15:00	06:20 14:50	40h 30m	40h 30m	5	1
22	07:15 15:45	RD	06:20 14:20	06:20 14:20	06:20 14:20	06:20 14:20	RD	32h 00m	40h 30m	4	1
23	RD	13:00 21:30	13:00 21:30	RD	RD	13:00 21:30	12:00 19:00	32h 30m	32h 30m	4	1
24	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
Cover	10 SP 4	15 SP 4	15 SP 4	15 SP 4	15 SP 4	15 SP 4	13 SP 4	35h 00m	38h 26m	4.40	16 one turn
Minutes	4,950	7,110	7,140	7,200	7,140	7,110	6,300	42,000	46,950		

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: distinct clock times Monday to Friday, **bold** where the week is one turn (one clock time Mon-Fri and one early-or-late family across every worked day — the page 1 figure) · Minutes: duty minutes per day, and Mon-Sat is the contract (20 × 35h = 42,000)

The words on this page

Link — the whole rota: 24 weeks laid out as 24 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 24 people.

Turn — a shift time, such as 06:20-13:45. An **early** starts before 09:00; a **late** starts after.

Cover (spare) week — a week with no fixed times. You work four duties of the seven days, placed by the roster clerk to cover leave and sickness.

The contract — 35 hours a week, averaged over the 24 lines; a cover week counts as a full contracted week.

Line — one week of the pattern, Sunday to Saturday. You move down a line each week, and line 24 goes back to line 1.

Rest day (RD) — a day off inside the contracted week. Two together make a proper break; one on its own does not count as a 48-hour break.

Sunday — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon-Sat and all week.

One-turn week — the same clock time all week (shown bold in the Turns column); the easiest kind of week to live around.

What was changed, and what could not be

One duty an hour shorter; everything else is the same work in a different order

Kept — the shape of a week

MEASURE	TODAY (20)	PROPOSED (24)
Working weeks that are one turn (one clock time Mon–Fri)	7 of 16	16 of 20
Weeks mixing early and late turns	7	0
Rest-day breaks that are two days or more	13 of 17	17 of 24
Single rest days between duties (not a 48h break)	4	7
Days worked in a week (lines × days)	1×3, 3×4, 11×5, 1×6	3×4, 16×5, 1×6
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	lines 6, 12, 18, 24
Distinct shift times (all but 11 on today's roster)	18	15
Mon–Sat hours, average week (the contract)	35h 00m	35h 00m
Sunday duties (not contracted; half the working lines)	8 of 16 lines	10 of 20 lines

Changed — the December 2026 headcount

PEOPLE ON DUTY	TODAY	PROPOSED	WHY
Monday to Friday	11–12	15	the contract: 20 working lines × 35h has to be worked somewhere
Saturday	10	13	owner's figure for December 2026, leaning late for events
Sunday	8	10	owner's figure for December 2026
Opening at 06:20 (07:15 Sunday) wk · Sat · Sun	4 · 4 · 3	3·5 · 4 · 4	four to open, every day — not met: Mon 5, Wed 3
Through to the close (23:55; 23:25 Sunday) wk · Sat · Sun	2 · 2 · 3	3 · 3 · 3	three to close, four on a Saturday — not met: Saturday 3
Still on duty at 22:00 wk · Sat · Sun	4 · 2 · 3	5 · 4 · 3	five from 22:00, Monday to Saturday — not met: Saturday 4

The duty table, in today's times

TURN	WK	SAT	SUN	WK	SAT	SUN
06:20–13:20 7h 00m · Early				1 +1		
06:20–13:25 7h 15m · Early	4			4		
06:20–13:45 7h 25m · Early	4			4		
06:20–13:50 7h 30m · Early				3 +3		
06:20–14:00 7h 40m · Early		4			4	
06:20–14:20 8h 00m · Early	2			1 -1	2 +2	
06:20–14:50 8h 30m · Early		3			2 -1	
07:00–15:00 8h 00m · Early				3 +3	2 +2	
07:15–15:45 8h 30m · Early			3			4 +1
08:00–16:20 8h 30m · Early	2			2		
08:30–16:30 8h 00m · Early			4			4
10:00–18:30 8h 30m · Early						1 +1
11:00–19:30 8h 30m · Late	4			4		
12:00–19:00 7h 00m · Late					1 +1	
12:00–20:00 8h 00m · Late		4			4	
13:00–21:00 8h 00m · Late			4			4
13:00–21:30 8h 30m · Late				4 +4	2 +2	
13:30–21:00 7h 30m · Late		4			4	
13:30–22:00 8h 30m · Late	4			4		
14:00–22:00 8h 00m · Late						2 +2
14:00–22:30 8h 30m · Late	2			2		
14:30–22:00 7h 30m · Late		2			2	
14:30–23:25 8h 55m · Late			3			3
14:45–23:55 9h 10m · Late		2			2	
15:00–23:00 8h 00m · Late				1 +1	1 +1	
15:15–23:55 9h 40m · Late	2			2		
15:25–23:25 8h 00m · Late						3 +3
15:55–23:55 8h 00m · Late					3 +3	
16:25–23:55 7h 30m · Late				3 +3		

Busiest weekday · Saturday · Sunday. The first three columns are what the 20-line link does now; the last three are the proposal. Green cells grew, amber shrank, a struck-through row is a time the proposal does not use.

Why the duties land where they do: Monday to Saturday totals exactly 42,000 minutes, which is 20 × 35h to the minute, so the contract is paid. But unlike a table built day-uniform, **the weekdays here are not equal** — Mon 7,110 · Tue 7,140 · Wed 7,200 · Thu 7,140 · Fri 7,110, and Saturday 6,300. Monday is the lightest day by some margin and Thursday the heaviest; that is a property of the design rather than a fault, but it is the first thing to put to the roster office, because it decides how much cover each day really has.

People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

How to read it: each blue cell is the number of people on duty in that hour (darker blue = more people). The orange row is the timetable — how much of the day's trains that hour carries (darker orange = busier). Read the columns top to bottom: cover should be thickest where the orange is darkest. The **fit** figure at the right end of each row scores that match for the whole day.

Monday to Friday window 06:20–23:55 · one row per distinct weekday

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY MON · WED		4	4	6	6	6	7	7	7	7	7	7	5	5	5	4	4	4	2	55.8
TODAY TUE · THU · FRI		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2	50
PROPOSED MON		5	7	7	7	7	7	7	10	8	5	8	8	8	8	8	8	5	3	27.6
PROPOSED TUE · THU		4	7	7	7	7	7	7	10	9	5	8	8	8	8	8	8	5	3	31.7
PROPOSED WED		3	6	6	6	6	6	6	10	10	6	9	9	9	9	9	9	5	3	43.9
PROPOSED FRI		4	7	7	7	7	7	7	10	8	5	8	8	8	8	8	8	5	3	27.8
DEC 2026 TRAFFIC	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66	

Saturday window 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2	45.7
PROPOSED		4	6	6	6	6	6	7	9	9	7	7	7	7	6	6	6	4	3	31.6
DEC 2026 TRAFFIC	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41	

Sunday window 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3	65.9
PROPOSED			4	4	4	5	5	5	5	7	10	6	6	6	5	5	5	3	3	43
DEC 2026 TRAFFIC			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50	

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. **Fit** is one number for how closely the people on duty follow the trains through the day: 0 would be a perfect match, lower is better, and today's link scores what it scores. (For the record: the squared gap between each hour's share of the day's traffic and its share of the cover, inside the day's window, measured on duty minutes — so an hour a duty only touches counts in proportion; the cells above count heads.) It is the measure the searched tables were chosen on, and the fit of the average weekday is the *Wk fit* on page 8. Sunday: 5 December 2026 timetable movements fall after the 23:25 finish — the standing question on whether Sunday's window moves; the window is stored per design, so this proposal can be rebuilt to either answer.

Reading it. The proposal raises the whole day rather than reshaping it, which is what the December 2026 timetable brief asked for — the station rosters posts (ticket office, gateline, passenger assist) plus break cover, and a post needs someone on it whether the hour carries 23 trains or 5. The weekday's two peaks are a duty length apart, so cover stays fairly flat between them by arithmetic, not neglect.

The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

Hard limits — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **6** days (today: 9)
Longest possible run is 6 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows. Worked as a BLOCK of four rather than split day-on-day-off, the answer is the same — **6** days — so this row does not depend on how a cover week is placed.
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)
Tightest anywhere: **14h 05m — line 9 Sat 15:55-23:55 into line 10 Sun 14:00-22:00** (today's tightest: 12h 30m). The generator refuses a design it cannot repair to this; this design was checked against it, not generated.
- ✓ **The contracted week, exactly** — **35h 00m** average Mon–Sat over the 24 lines, cover weeks counted as contracted weeks
700h of duty a week across 20 working lines. Sundays (82.50h) sit on top as RDW, as they do today. Individual weeks range 30h 00m–42h 30m; only the average is the contract.

December 2026 timetable design figures — the staffing shape agreed for the new timetable

RULE	PROPOSAL	
Four on at the open, every day	× 3–5 weekday · 4 Saturday · 4 Sunday	
Three through to the close; four on a Saturday	× 3 weekday · 3 Saturday · 3 Sunday	
Five still on duty at 22:00	× 5 weekday · 4 Saturday	a 22:00 finish is not "on at 22:00"
Fourteen on a Saturday, ten on a Sunday	× 13 · 10	
Cover weeks, evenly spread	✓ 4 weeks at lines 6, 12, 18, 24 — gaps 6, 6, 6	
About 4.2 days a week worked, Mon–Sat	× 4.40 over the 20 working lines	
No :05 or :10 times except the open and close	✓ none	
Late turns slightly shorter than most earlies	× not met — lates 7h00–8h30, earlies 7h00–8h30	the longest late is longer than the longest early

To settle before it is frozen

Both gates are clear and the staffing is untouched. 35h to the minute, no rest under 12h, longest run 6 — inside the design target of 7, not just the 13-day limit — and the cover weeks still perfectly even. One ORR factor remains present: FF19, a single start-time jump over two hours. FF15 is clear at 4. **What could NOT be repaired by rearranging.** The late-shorter-than-early lever is a property of the duty TABLE, not of the order: earlies and lates here both run 7h00–8h30, so no arrangement can make one strictly shorter than the other. Meeting it needs a different table, which is the trade *By the Book* makes and this design does not. **Sunday's finish** — five December 2026 timetable movements fall after 23:25; this design inherits today's window rather than deciding it.

ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021). $\triangle$ present — the pattern is in this design and worth a look, not a breach · $\checkmark$ clear — it is not · $\bullet$ standing — true of the station itself, not of any design · — does not apply here. Nathan assesses against this list.

The Office of Rail and Road lists 25 roster patterns that tend to tire people — long runs of earlys, short gaps between duties, start times that jump about. For each one this table asks whether the pattern is in the rotation, today and proposed, and how big it is. The list is guidance: a factor present is a question for the room, never a pass or a fail, and a design showing nothing is not thereby approved.

CODE	FACTOR	TODAY'S LINK	PROPOSED
FF1	Night shift covering 00:00–05:00 Time of day	--	--
FF2	Early shift starting 05:00–07:00 Time of day	$\bullet$ 24	$\bullet$ 24
FF3	Very early shift starting before 05:00 Time of day	$\checkmark$ 0	$\checkmark$ 0
FF4	Very early shift (before 05:00) over 8h Duty length	--	--
FF5	Day shift over 12h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF7	Early shift over 10h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF8b	Fewer than 2 days rest after a block of early starts Recovery time	$\triangle$ 3	$\checkmark$ 0
FF11	More than 13 consecutive shifts without a 48h break Recovery time	$\checkmark$ 12 $\checkmark$ 9 as rostered	$\checkmark$ 11 $\checkmark$ 10 as rostered
FF10	More than 4 consecutive 12h day shifts Cumulative	$\checkmark$ 0	$\checkmark$ 0
FF15	More than 4 consecutive early shifts in a rotating pattern Cumulative	$\triangle$ 6	$\checkmark$ 4
MRSF	More than 12 consecutive day shifts Cumulative	$\checkmark$ 9	$\checkmark$ 6
MRSF	More than 55 hours worked in any 7-day period Cumulative	$\triangle$ 60.8	$\checkmark$ 50
MRSF	More than 7 consecutive 8h shifts (definition to confirm) Cumulative	$\triangle$ 9	$\checkmark$ 6
MRSF	More than 6 consecutive night or early shifts in a permanent pattern Cumulative	--	--
FF17	Backward rotating pattern (definition to confirm) Circadian	$\checkmark$ 10 backward / 12 forward	$\checkmark$ 11 backward / 12 forward
FF18	Rotating pattern of about a week (definition to confirm) Circadian	$\bullet$ 20-line rotation · typically 4h 0m a week	$\bullet$ 24-line rotation · typically 3h 27m a week
FF19	Successive start times varying by more than 2 hours (definition to confirm) Circadian	$\triangle$ 6	$\triangle$ 1
FF6	Night shift over 10h Night shifts	--	--
FF8	Less than 2 days rest after consecutive nights Night shifts	--	--
FF9	Less than 14h rest in any 24h for night shifts Night shifts	--	--
FF12	Only one day rest after night shifts Night shifts	--	--
FF14	More than 4 consecutive nights in a rotating pattern Night shifts	--	--
FF16	More than 3 consecutive night shifts over 8h Night shifts	--	--
FF20	First night shift Night shifts	--	--
FF13	Less than 12h rest in any 24h Intervals between duties	$\checkmark$ 0	$\checkmark$ 0

Present: 1 (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. **Two readings of a cover week, on the one row they move.** A cover week is worked four days of seven and the link does not say which four, so a run figure is a range. The headline number is the ceiling — the four split day-on-day-off, which supplies no 48-hour break and joins the blocks either side. It is reachable: of the 35 ways to place four duties in seven days, the 10 that leave no two rest days together produce exactly it. Worked as a BLOCK, which is what a cover week looks like on the roster, the three rest days fall together and the week always supplies a break — that is the *as rostered* figure beneath it. Checked on every row: FF11 is the only one where the two differ. Which reading applies is a question for the roster office, not for this sheet. This sheet is an aid to a conversation, not a fatigue risk assessment.

How it was repaired

Reproducible: the same starting grid, seed and objective give the same rotation on any machine

In plain terms. Nobody searched for this design. It was typed in from the table exactly as supplied and the app's own rule modules measured it; nothing below was chosen by a computer. Everything on these pages can be re-checked by pasting page 9 into the Links workspace.

How it was repaired

1 • One duty, and it fixed both gates. The supplied grid was 60 minutes over the contracted week and carried one rest of 11h15 — line 7's Saturday 12:00–20:00 into line 8's Sunday 07:15–15:45. That turn is the only place 12:00–20:00 appears anywhere, and it sits in both failures. Shortening it to **12:00–19:00** removes exactly the 60 surplus minutes and lifts that rest to 12h15. Saturday cover at 19:00 goes from seven to six; nothing else moves.

2 • Then the shape was searched, not the staffing. Two moves only: swap two lines' duty within one day column, or swap two whole lines. Both leave each day's duties exactly as they were, so coverage,

headcounts, the duty table and the contracted week are invariant *by construction* — the hour-by-hour heat map is identical to the supplied design's, and the search asserts that rather than assuming it. 60,000 candidates per restart, four restarts, four seeds, then a steepest-descent polish of every same-day swap and every pair of lines.

3 • The judge is the workspace's own — runDesignChecks, assessFatigue, assessHardLimits and scoreOrder, under anneal.mjs's measured weights. Four seeds gave four answers, all clearing every hard gate; this one is kept because it alone also clears FF15.

Alternatives measured

DESIGN	RUN	FF PRESENT	WKENDS	ONE - TURN	STEP	WK FIT	SCORE
FT-24-R21 · b76bf9e1 — Fifteen Turns Repaired (this proposal)	6	1	6	16/20	3h 27m	30.7	—
Same Turns · d15e1b74 — the searched proposal	6	1	6	18/20	2h 33m	46.3	—
By the Book · 0f14abce — the searched proposal	6	0	6	12/20	2h 52m	30.8	—
FT-24-R7 — seed 7, same search · 5c4e9308	6	2	7	14/20	2h 8m	30.7	—
FT-24-R13 — seed 13, same search · 1e875350	6	2	6	16/20	3h 46m	30.7	—
FT-24-R34 — seed 34, same search (lowest blended score) · f3530c54	6	2	6	17/20	2h 21m	30.7	—
FT-24-EXT — the design AS SUPPLIED, before repair · 9a028392	9	7	2	12/20	2h 16m	30.7	—
Workspace default · 29ec9810 (Dec 2026 table, generated)	7	5	5	0/20	4h 16m	30.8	—
Today's 20-line link (for scale)	9	5	4	7/16	4h 0m	44.7	—

How this one was picked: every row above clears all four hard gates and every one keeps the coverage curve identical — so the choice was between advisory outcomes, not safety. R21 is the only seed that also clears FF15, leaving a single factor present; R34 scores lower on the blended objective through a gentler 2h21 step, and R7 gives one more full weekend off. The blended score was NOT taken as the answer: fewer factors present is what the ORR panel reports and what a reader acts on. *Wk fit* is how evenly the weekday cover follows the December 2026 timetable traffic curve, lower being more even.

LOAD IT

The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

Links → Import, then paste the whole block below. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied beside this PDF as Fifteen-Turns-Repaired-FT-24-R21-import.txt (tab-separated, pastes directly) and Fifteen-Turns-Repaired-FT-24-R21.json (the app's own format).

1	07:15-15:45	06:20-13:50	RD	RD	06:20-13:50	07:00-15:00	06:20-14:20
2	07:15-15:45	06:20-13:20	RD	06:20-13:20	06:20-13:20	06:20-13:20	06:20-14:50
3	RD	RD	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	15:00-23:00
4	RD	06:20-13:50	06:20-13:50	RD	RD	06:20-13:50	07:00-15:00
5	07:15-15:45	06:20-13:50	06:20-13:50	RD	RD	06:20-13:50	07:00-15:00
6	SP	SP	SP	SP	SP	SP	SP
7	15:25-23:25	16:25-23:55	RD	16:25-23:55	16:25-23:55	16:25-23:55	RD
8	RD	14:00-22:30	14:00-22:30	14:00-22:30	14:00-22:30	15:00-23:00	RD
9	RD	13:00-21:30	13:00-21:30	13:00-21:30	13:00-21:30	RD	15:55-23:55
10	14:00-22:00	14:00-22:30	14:00-22:30	14:00-22:30	RD	RD	15:55-23:55
11	15:25-23:25	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	RD	RD
12	SP	SP	SP	SP	SP	SP	SP
13	14:00-22:00	13:00-21:30	RD	13:00-21:30	14:00-22:30	14:00-22:30	RD
14	RD	07:00-15:00	07:00-15:00	07:00-15:00	07:00-15:00	07:00-15:00	RD
15	RD	06:20-14:20	06:20-13:20	06:20-13:50	06:20-13:50	RD	RD
16	RD	07:00-15:00	07:00-15:00	07:00-15:00	07:00-15:00	RD	06:20-14:20
17	15:25-23:25	16:25-23:55	16:25-23:55	RD	RD	16:25-23:55	15:55-23:55
18	SP	SP	SP	SP	SP	SP	SP
19	10:00-18:30	RD	RD	13:00-21:30	13:00-21:30	13:00-21:30	13:00-21:30
20	RD	RD	13:00-21:30	13:00-21:30	13:00-21:30	13:00-21:30	13:00-21:30
21	RD	RD	07:00-15:00	07:00-15:00	07:00-15:00	07:00-15:00	06:20-14:50
22	07:15-15:45	RD	06:20-14:20	06:20-14:20	06:20-14:20	06:20-14:20	RD
23	RD	13:00-21:30	13:00-21:30	RD	RD	13:00-21:30	12:00-19:00
24	SP	SP	SP	SP	SP	SP	SP

Prepared 18 September 2026 · re-rendered 24 September 2026. The code **b76bf9e1** in every footer is the design's fingerprint — eight characters computed from these 168 cells and nothing else. If the workspace shows the same code after you paste, you have exactly this design; if one cell differs, the code changes. Every figure in this document was computed by the Marylebone Roster app's Links modules (v24.22) from exactly these cells; import them and the workspace's Design checks, hard limits, fatigue factors and coverage cards will restate them.